

Exercise 1: Setting Up JUnit

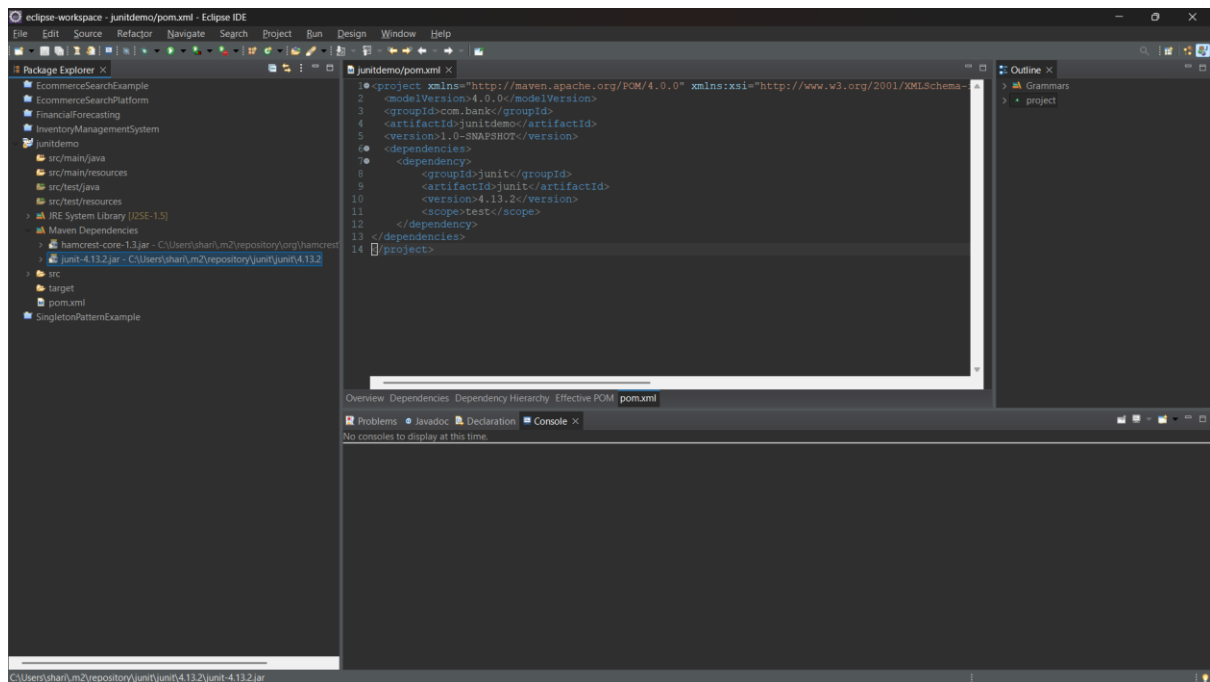
JUnit

- JUnit is a Java testing framework used for unit testing.
- It helps developers test individual units of code automatically.
- JUnit improves code quality and simplifies debugging.

Dependency Used

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>4.13.2</version>
  <scope>test</scope>
</dependency>
```

OUTPUT

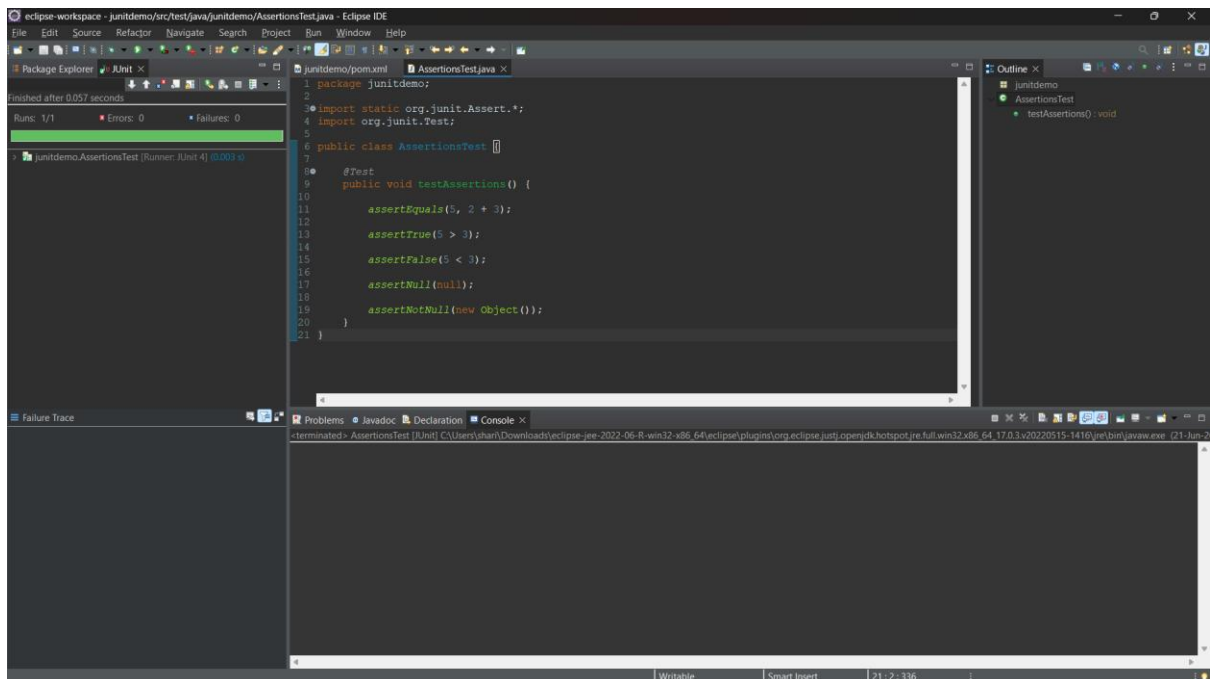


Exercise 3: Assertions in Junit

AssertionsTest.java

```
package junitdemo;
import static org.junit.Assert.*;
import org.junit.Test;
public class AssertionsTest {
    @Test
    public void testAssertions() {
        assertEquals(5, 2 + 3);
        assertTrue(5 > 3);
        assertFalse(5 < 3);
        assertNull(null);
        assertNotNull(new Object());
    }
}
```

OUTPUT



Exercise 4: (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

Arrange: Prepare the test data and required objects before executing the test.

Act: Execute the method or operation that needs to be tested.

Assert: Verify that the actual result matches the expected result using JUnit assertions.

@Before (Setup Method): The @Before annotation is used to initialize test data and resources before each test case is executed.

@After (Teardown Method): The @After annotation is used to clean up resources and reset the test environment after each test case execution.

Result: The test case executed successfully using the AAA pattern, with setup and teardown methods running before and after the test, resulting in **Runs: 1, Failures: 0, Errors: 0**.

OUTPUT

The screenshot displays the Eclipse IDE interface during a JUnit test run. The central editor shows the source code of `AAATest.java`, which implements the AAA pattern:

```
1 package junitdemo;
2
3 import static org.junit.Assert.*;
4
5 import org.junit.After;
6 import org.junit.Before;
7 import org.junit.Test;
8
9 public class AAATest {
10
11     private int number;
12
13     @Before
14     public void setUp() {
15         System.out.println("Setup Method Executed");
16         number = 10;
17     }
18
19     @After
20     public void tearDown() {
21         System.out.println("Teardown Method Executed");
22     }
23
24     @Test
25     public void testAddition() {
26         // Test logic would go here
27     }
28 }
```

The left sidebar shows the Package Explorer with the `JUnit` tab active, indicating a successful run of `testAddition` (0.001 s). The bottom console window displays the output of the test, showing the execution of the `setUp` and `tearDown` methods.